Public Spending and Social Outcomes in the EU

Introduction

For this project, we decided to look at **public spending on health and education in the European Union** (EU), and **whether countries that spend more tend to have better social outcomes**. To measure this, we choose **two outcomes**: the *poverty rate* (the share of people living below 60% of the median income) and *life expectancy at birth*. Our data comes directly from Eurostat and covers 24 EU member states from 2016 to 2023. The EU is an interesting case in the sense that it includes two quite different groups of countries: the EU-15 are the older member states (UK excluded) and the post-2004 group (joined mainly from Central and Eastern Europe). These two groups have different income levels and welfare state traditions, so comparing them appeared to be a robust starting point on which better develop the analysis.

Our three questions for this case-study are the following:

How different is public spending between the two groups?
Is higher spending associated with lower poverty and higher life expectancy?
What are the overall trends between 2016 and 2023?

Data source (Eurostat)

We downloaded three datasets directly from the data browser affiliated to Eurostat. Government spending, more specifically on health (COFOG code GF07) and education (GF09) comes from GOV_10A_EXP, with filters applied to only consider the general government (sector S13) and expressed as a percentage of GDP. The poverty rate comes from ILC_LI02, which measures the share of the population with income below 60% of the national median. Life expectancy at birth comes from DEMO_MLEXPEC, using the combined male and female figure as per the expectancy at the very first year of life. Three countries (Germany, France and Cyprus) were missing from our spending download despite explicitly including them in the downloading form, so we worked on 24 out of 27 EU member states. The panel runs from 2016 to 2023, giving us 211 country-year observations after dropping rows with missing values. We also created a total spending variable by adding health and education spending together.

However, this spending file includes multiple entries per countries (absolute values, per capita figures and percentages of GDP), causing inconsistencies. To resolve this we filtered the data to maintain only rows where *unit == "PC_GDP",* doing so each country appeared only once in the merged dataset.

Results

Firstly, we were able to observe descriptive statistics. To present them efficiently, we chose to use a table showing summary statistics for the dataset (24 countries between 2016 and 2023) with the output of df.describe() and the group means of df.groupby("group") in the terminal:

Variable	Mean	Deviation	Min	Max

Health Spending (% of GDP)	6.48	1.39	3.60	10.10
Education Spending (% of GDP)	4.92	1.02	2.50	7.80
Total Spending (% of GDP)	11.40	2.04	7.20	15.60
Poverty Rate (%)	16.67	3.97	8.60	25.30
Life expectancy at birth (in years)	79.92	3.00	71.20	84.00

From it, we find that average total spending is 11.4% of GDP, ranging from 7.2% in Romania to 15.6% in Denmark. The poverty rate averages 16.7% and varies a lot across countries, with a standard deviation of almost 4 percentage points. When we split by group, EU-15 countries spend on average 12.1% of GDP versus 10.7% for post-2004, which represents a gap of 1.4 percentage points. The difference in life expectancy is more striking: 82.2 years on average for EU-15 versus 77.6 for post-2004, therefore a gap of almost 5 years.

We can now use the plots generated by our code. The very first chart we need is a histogram of the total spending by group between 2016 and 2023:

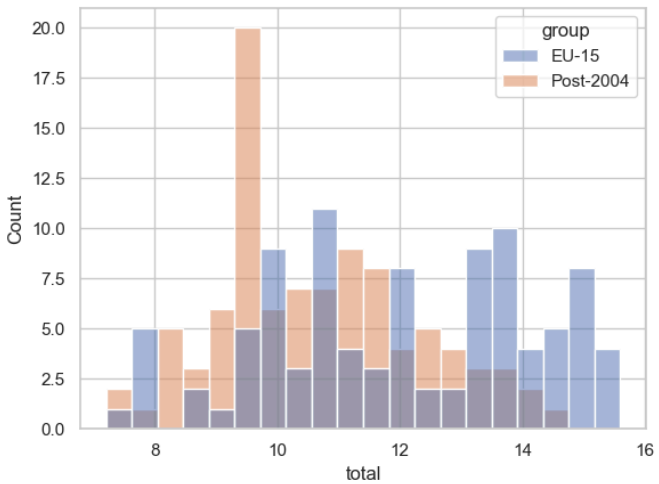

Chart 1: Histogram of total spending by EU group (2016 to 2023)

Chart 1 shows that the two distributions overlap more than expected: even if the gap between the two groups is visible, it is less strong than the life expectancy difference might have suggested.

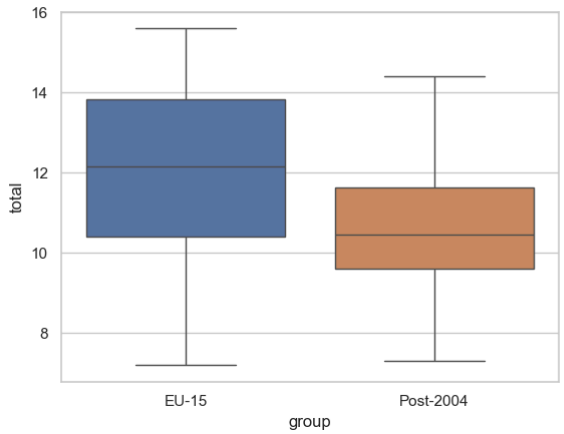

Chart 2: Total spending by EU group: boxplots (2016 to 2023)

More specifically, we then used boxplots (Chart 2), showing total spending by group. Chart 2 confirms this: the two boxes overlap, meaning some post-2004 countries spend as much as the lower-end EU-15 countries. However, the EU-15 median and upper range are clearly higher, which is consistent with the 1.4 percentage point gap in group averages.

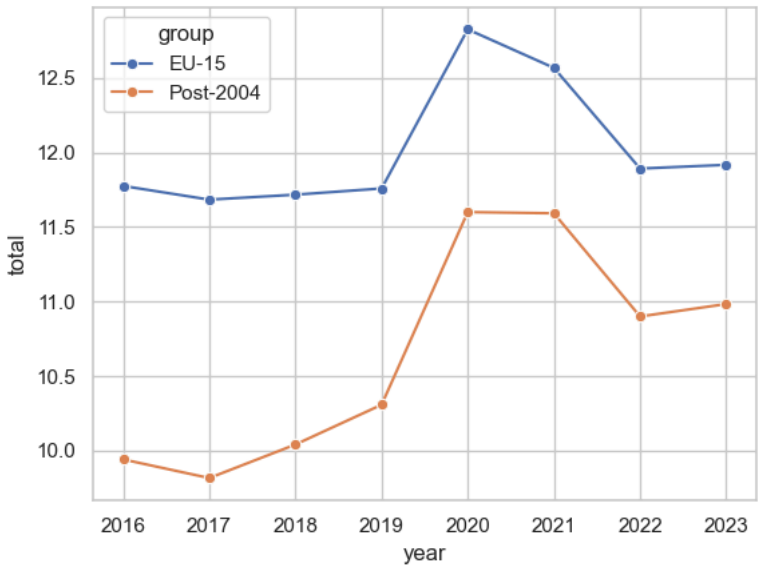

Chart 3: Average total spending by group over time (2016 to 2023)

From this point, we can start to give an idea of the trends at work among countries. In that sense, Chart 3 shows group averages of total spending over time. The Covid effect is quite significant, with health spendings that jumped in both groups in 2020, and life expectancy dropping sharply in 2020 and 2021 before going up again. Indeed, there was a pandemic. But beyond this, total spending was relatively stable across the period, with EU-15 countries consistently spending more throughout the years. At the same time, the poverty rate had already been declining in both groups before 2020.

Now, if we rank all countries by total spending in 2023 to observe above the group argument, we observe the following:

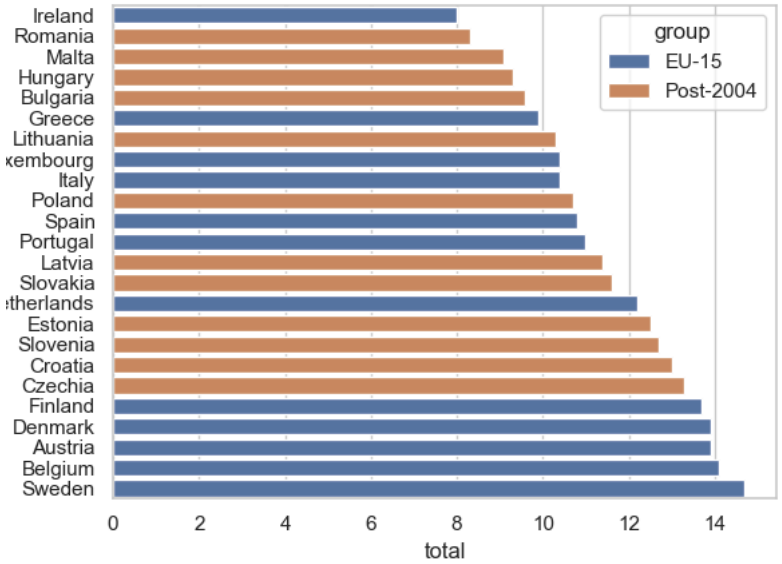

Chart 4: Total spending by country, 2023

Chart 4 shows that Denmark, Finland, Belgium, and Sweden are at the top, all above 13% of GDP. Romania, Malta and Bulgaria are at the bottom. The chart also shows that the ranking is relatively stable year-on-year, the 2023 snapshot being representative of the full period.

This allows us to then use correlations in order to have an idea of the outcomes of these spendings among countries and groups:

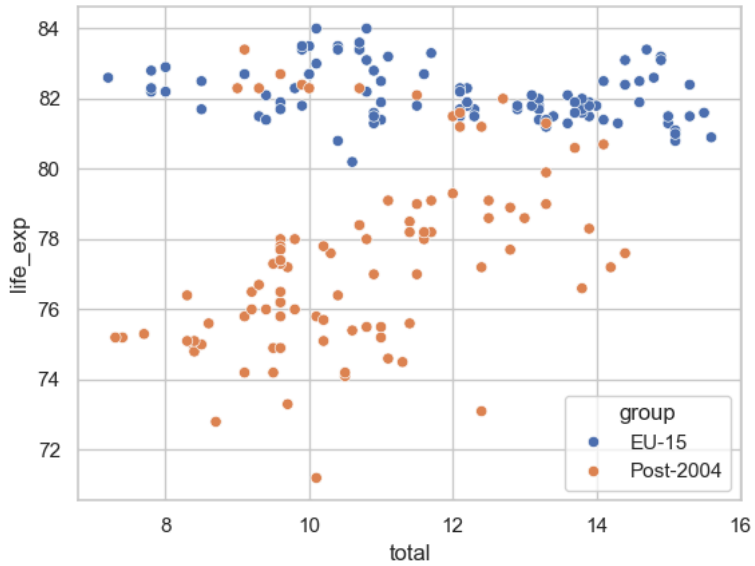

Chart 5: Total spending vs life expectancy (2016 to 2023)

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Chart 5 plots total spending against life expectancy for all country-year observations, coloured by group. The correlation with poverty (r = -0.45) suggests that countries that spend more tend to have lower poverty rates. The relationship with life expectancy is positive but also weaker (r = +0.35).

Furthermore, we should be careful when interpreting these numbers. Countries that spend more on health and education are also generally wealthier, so we cannot really separate the effect of spending from the effect of overall wealth. Still, the direction of the correlations matches our expectations, and the relationship with poverty is reasonably strong.

Conclusion

Three elements came out of this analysis.

First, on average, EU-15 countries spend more on health and education on average (12.1% of GDP vs 10.7%), though the gap is more moderate than the life expectancy difference of nearly 5 years might suggest.

Second, higher spending is associated with lower poverty (r = -0.45) and somewhat higher life expectancy (r = +0.35), but we cannot claim this is a causal effect.

Third, COVID had a visible and measurable impact on both spending and outcomes in 2020-2021, which shows up clearly in the time series.

The main limitation of our project is that the five charts produced are individually focused, and the code we design generates one variable per chart rather than multi-panel figures, which makes direct side-by-side comparison slightly harder to read.